instead of "ü". Rails takes a number of internal steps to mitigate common causes of these problems that can be automatically detected and corrected. However, if you have external data that is not stored as UTF-8, it can occasionally result in these kinds of issues that cannot be automatically detected by Rails and corrected.

Two very common sources of data that are not UTF-8:

- Your text editor: Most text editors (such as TextMate), default to saving files as UTF-8. If your text editor does not, this can result in special characters that you enter in your templates (such as é) to appear as a diamond with a question mark inside in the browser. This also applies to your i18n translation files. Most editors that do not already default to UTF-8 (such as some versions of Dreamweaver) offer a way to change the default to UTF-8. Do so.
- Your database: Rails defaults to converting data from your database into UTF-8 at the boundary. However, if your database is not using UTF-8 internally, it may not be able to store all characters that your users enter. For instance, if your database is using Latin-1 internally, and your user enters a Russian, Hebrew, or Japanese character, the data will be lost forever once it enters the database. If possible, use UTF-8 as the internal storage of your database.